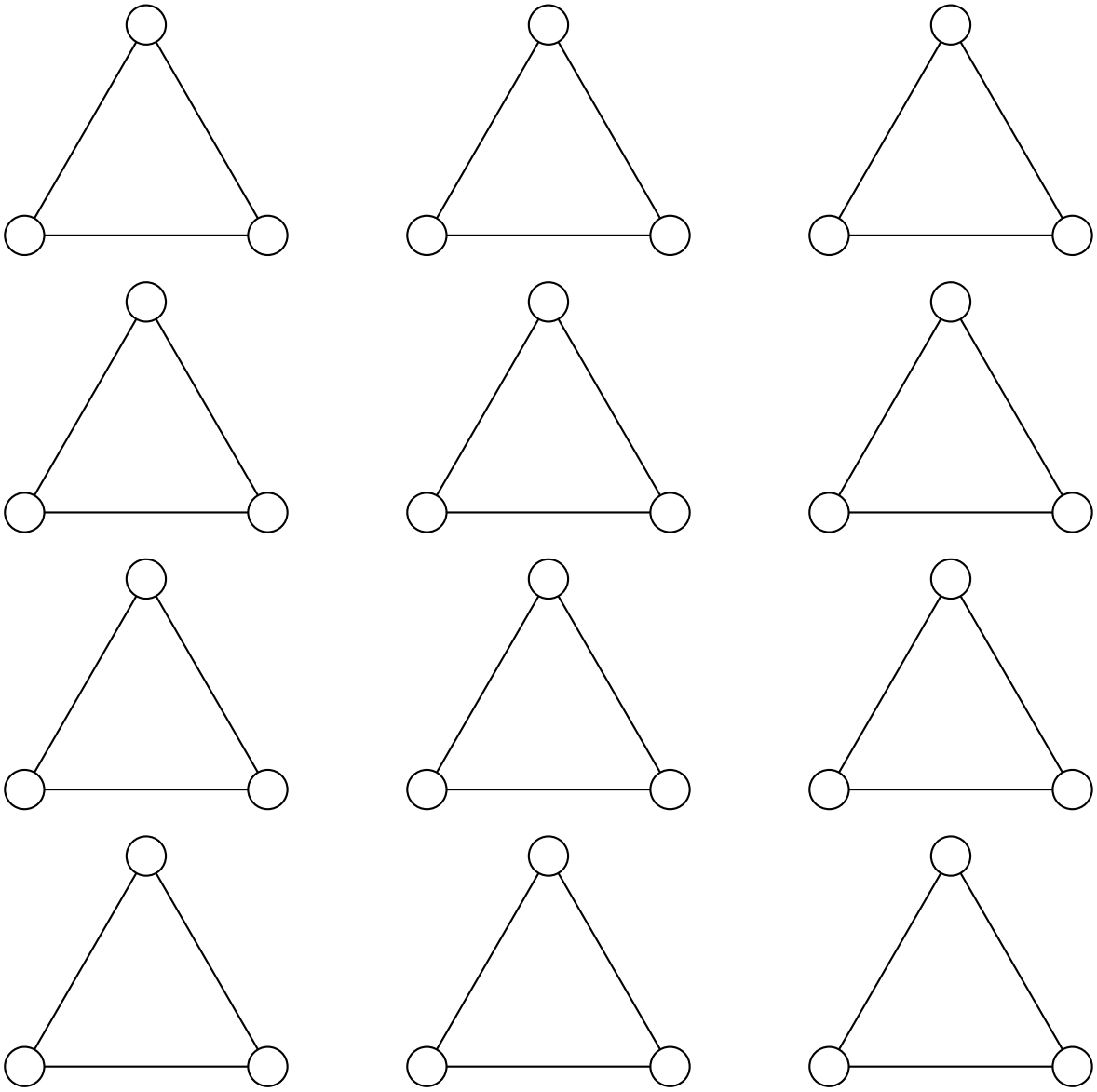
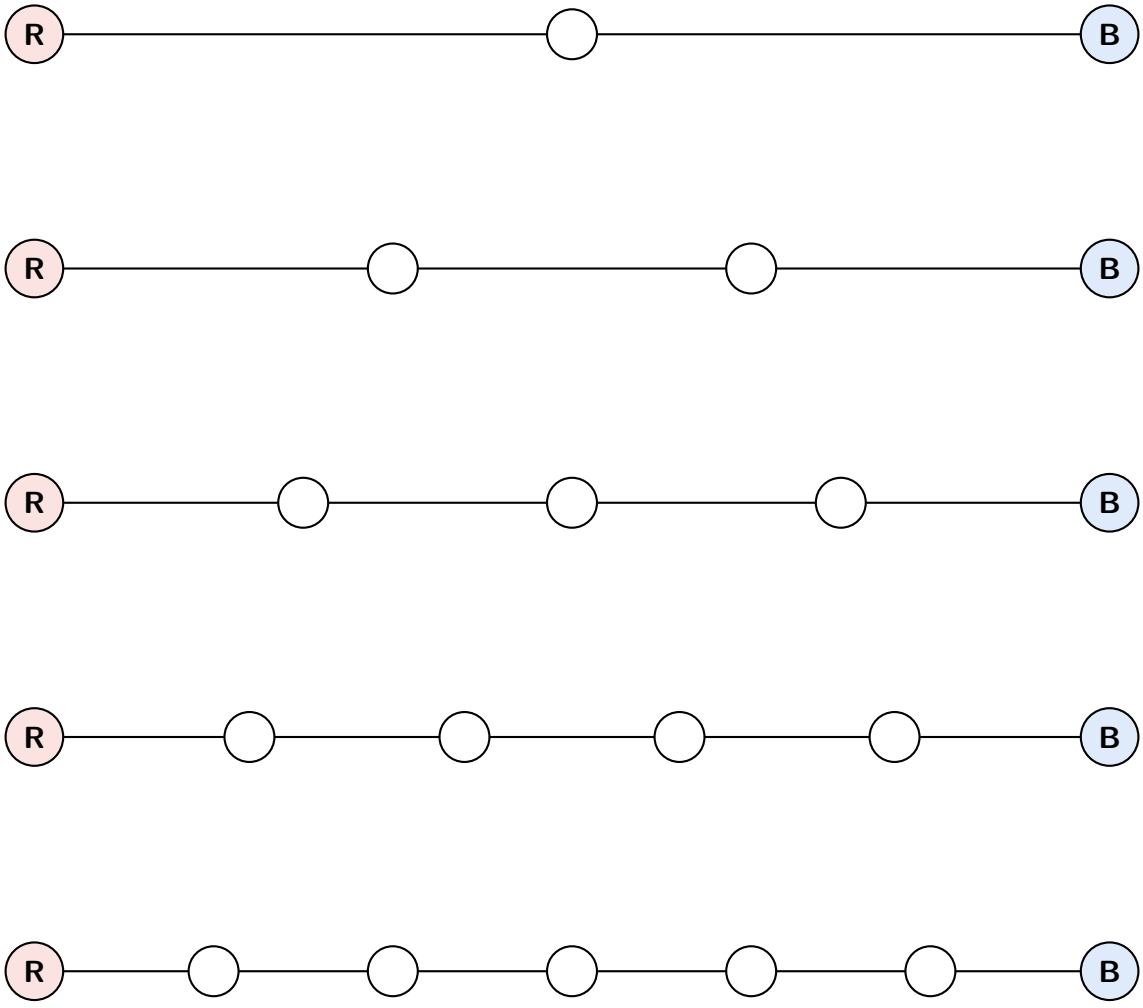
Write R, B, or Y on every empty dot. Keep the printed letters. Each outside side allows only the letters beside it. A little triangle has no lines inside.

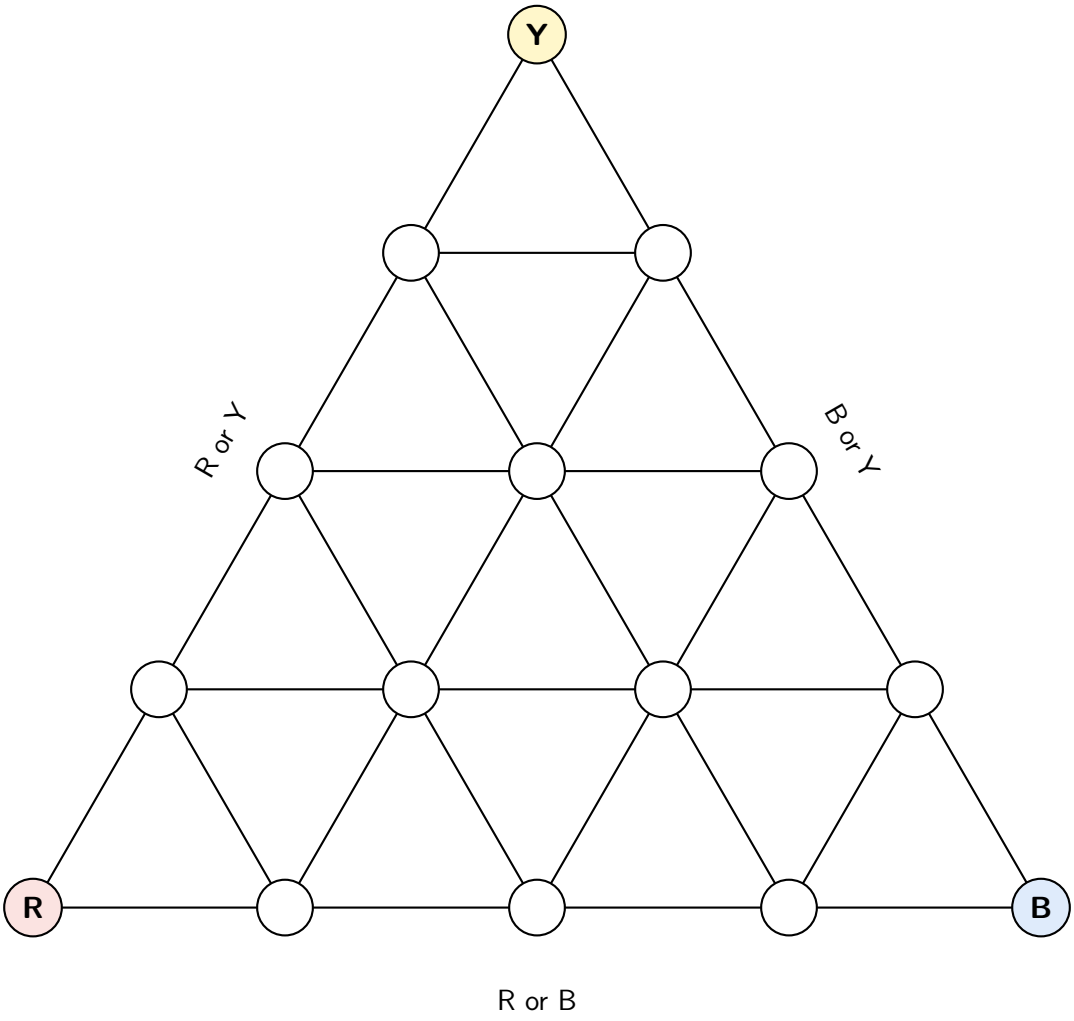
**Problem 3:** Put R, B, or Y on each dot. Find every different filling, counting turns and flips as the same. Mark the doors in each filling. Could a triangle have three doors?



**Problem 4:** Write R or B on every empty dot and mark the doors. Can any row have an even number of doors? Explain why or why not.



**Problem 5:** Fill the dots so that exactly three little triangles have R, B, and Y. Draw all routes that start at outside doors. Can all those routes end outside?



**Problem 6:** Can the dots be filled so that no little triangle has R, B, and Y? Give a reason that covers every filling allowed by the side rules.

